

7.8 NERVOUS SYSTEM AND SENSE ORGANS OF PERIPLANETA

The nervous system of cockroach consists of central nervous system, peripheral nervous system and autonomous nervous system.

7.8.1 Nervous System

I. Central Nervous System

It consists of a **nerve ring**, and a **ganglionated double ventral nerve cord**.

1. Nerve ring

The nerve ring, which is present around the oesophagus, is formed by the following.

A. Brain

Brain lies above the oesophagus. The brain is mainly a sensory and an endocrine centre. Three lobes of the brain are **protocerebrum**, **deutocerebrum** and **tritocerebrum**. The protocerebrum receives sensory impulses from the compound eyes through optic nerves; deutocerebrum receives sensory impulses from antennae through antennal nerves; and tritocerebrum receives sensory impulses from the labrum. Hence brain is principally '**sensory**' in nature.

B. Sub-oesophageal ganglion

It lies below the oesophagus. It is the principal *motor center*, that controls the movements of **mouthparts**, **legs** and **wings**. It is formed by the fusion of **paired ganglia** of mandibular, maxillary and labial segments of the head.

C. Circum-oesophageal connectives

A pair of circum-oesophageal connectives is present around the oesophagus, connecting the tritocerebrum with the sub-oesophageal ganglion/ sub oesophageal ganglia.

2. Ventral nerve cord

The two ventral nerve cords are solid and ganglionated. They arise from the sub-oesophageal ganglion and extend upto the 7th abdominal segment. The two

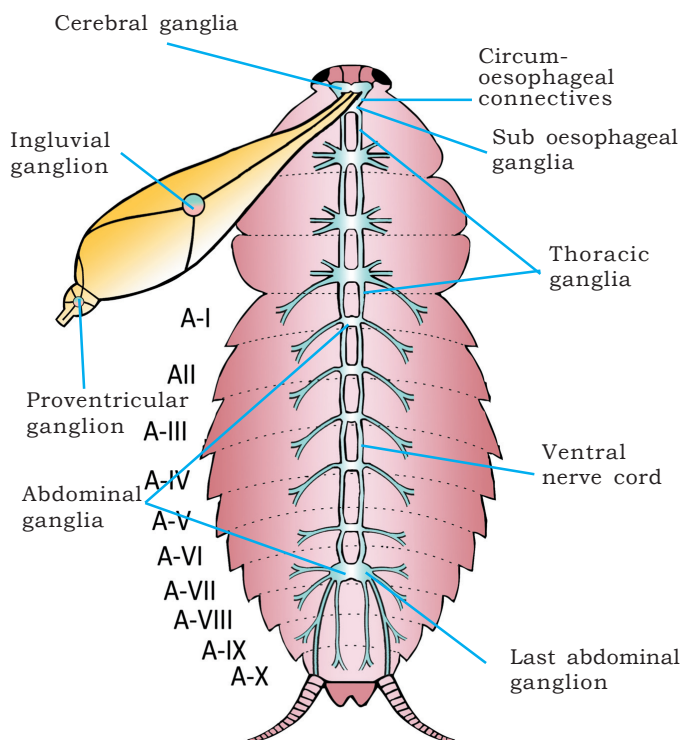


Fig. 7.10 *Periplaneta* : Nervous system

nerve cords remain separate except at the ganglia. Three thoracic ganglia are present, one in each thoracic segment. In addition, there are six abdominal ganglia. The first to the fourth abdominal segments have one abdominal ganglion each. The 5th abdominal segment has no ganglion. The serially 5th abdominal ganglion is present in the 6th segment. The serially 6th abdominal ganglion is present in the 7th segment. The last or the 6th abdominal ganglion is the *largest* of all the abdominal ganglia. It is formed by the fusion of the ganglia of the 7th, 8th, 9th, and 10th abdominal segments.

II. Peripheral Nervous System

It consists of nerves arising from the central nervous system. It receives a pair of, **optic nerves**, from the compound eyes, a pair of **antennal nerves**, from the antennae and a pair of **labral nerves**, from the labrum. Motor neurons of the frontal nerve to the **frontal ganglion** join the sensory neurons of the labral nerve to form the **labro-frontal nerve** arising from the tritocerebrum. Sub-oesophageal ganglion gives off motor nerves to the mandibles, maxillae, labium, wings and legs. It is the principal 'motor centre' in the body.

Thoracic ganglia supply nerves to the parts of their respective segments. Metathoracic ganglia send nerves to the first abdominal segment also. Nerves from the first four abdominal ganglia supply to the organs of the segments 2-6 serially (the 1st to the 4th ganglia innervate segments 2nd to 5th respectively). The 5th ganglion present in the 6th segment innervates the organs of the 6th segment. All organs present in 7th to 10th segments receive nerves from the last abdominal ganglion (present in the 7th segment). The organs include the reproductive organs, copulatory appendages besides anal cerci.

If the sub-oesophageal ganglion is damaged hypothetically, which parts of the cockroach are affected?

III. Autonomous Nervous System

This system is also called **stomatogastric nervous system** or 'visceral nervous system'. It controls the visceral organs, particularly the muscles of the alimentary canal, and the heart. Autonomous nervous system includes four ganglia, a **frontal ganglion** on the dorsal wall of the pharynx, in front of the brain, **hypocerebral ganglion** or occipital ganglion above the oesophagus, behind the brain, a **visceral ganglion** or **ingluvial ganglion** on the wall of the crop and a **proventricular ganglion** on the gizzard. These ganglia contain the 'somata' of the post ganglionic motor neurons. Preganglionic motor neurons of tritocerebrum go to the frontal ganglion as labro-frontal and frontal nerve. Frontal ganglion is connected to the hypocerebral ganglion by a '**recurrent nerve**'. Hypocerebral ganglion is connected to the visceral ganglion and in turn the visceral ganglion is connected to proventricular ganglion.

The movements of which organs in cockroach are controlled by its autonomous nervous system?

7.8.2 Sense organs

In cockroach, various structures like antennae, labrum, maxillary palps, labial palps, compound eyes, simple eyes (**fenestrae**) etc., are sensory in nature. These structures contain various receptors like chemoreceptors, mechanoreceptors, photoreceptors, thermoreceptors, etc. Simple eyes and **ommatidia** of a compound eye are photoreceptors. **Sensillae** are the units of cuticular receptors and chemoreceptors. **Scolopidia** are the subcuticular units of mechanoreceptors of **chordo-tonal organs** (**Johnston's organs** in the antennal pedicels and flagella, **subgenual organs** on the proximal parts of tibiae and **tympanal organs** on the anal cerci; they are sensitive to movement of flagella of antennae, ground vibrations and sound vibrations respectively).

I. Compound eye

A pair of dark kidney shaped compound eyes is present on the dorsolateral sides of the head, one on each side. Each compound eye is composed of about 2000 functional units called **ommatidia**, which are optically independent. The outer surface of the compound eye is divided into about 2000 hexagonal areas called **facets**. Each facet is the outermost portion of a corresponding ommatidium.

1. Structure of an Ommatidium

Each typical ommatidium is an elongated sub unit of the compound eye, consisting of the following parts.

1. **Cornea:** It is the outermost part of an ommatidium and corresponds to a 'hexagonal facet' of the compound eye. It is a biconvex, transparent part of the cuticle and allows light to pass through it. Cornea is secreted by specialized **corneagen cells** of the epidermis. Cornea is a 'refractive' region of an ommatidium.
2. **Corneagen cells or lenticular cells:** These are two transparent specialized epidermal cells that secrete cornea.

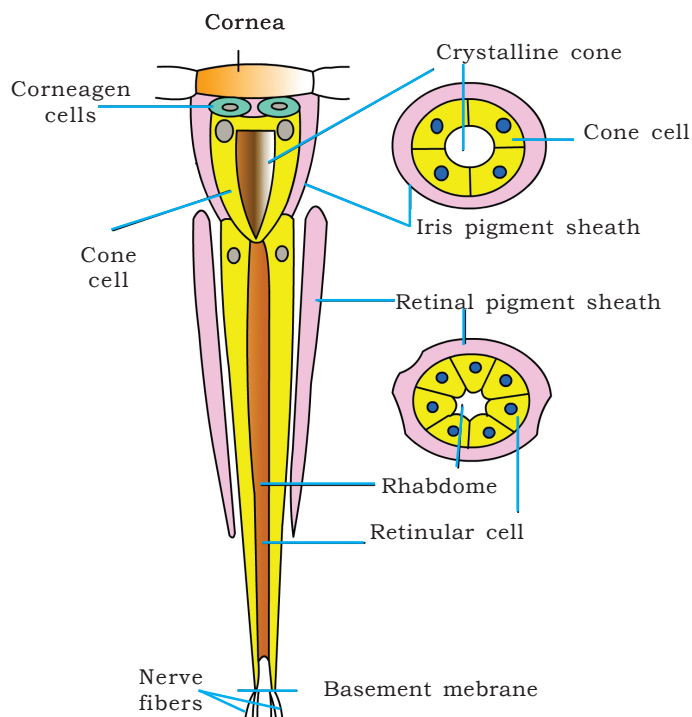


Fig. 7.11 Structure of a typical ommatidium

3. **Vitrellae or cone cells (Semper cells):** These are four transparent more or less conical cells that lie below the corneagen cells. They surround the transparent crystalline cone. Crystalline cone is secreted by the cone cells.
4. **Crystalline cone:** It is the transparent conical structure that is secreted by the vitrellae and is surrounded by them. Light absorbing dark primary pigment cells surround the vitrellae. The region containing the cornea and crystalline cone constitute the **dioptrical** or 'focussing region' of the ommatidium. Crystalline cone focuses the light on to the next part of the ommatidium.
5. **Retinulae:** These are the innermost and elongated cell of an ommatidium. They are seven in number. They rest on the basement membrane. Each cell bears microvilli towards the inner surface. Microvilli of each retinular cell collectively form a **rhabdomere** that contains photoreceptor pigments. These rhabdomeres fuse along the axis of the ommatidium to form the **rhabdome** in the centre. Retinulae are the nerve cells from which sensory nerve fibers leave as the optic nerve to the protocerebrum. They are the **photoreceptor** cells of the ommatidium. Rhabdome and retinulae form the retinal or receptor region. Receptor region is surrounded by seven secondary pigment cells, which absorb light and serve to isolate each ommatidium from the rest (retinal pigment sheath).

2. Functioning of an ommatidium: The position of retinulae below the vitrellae or cone cells, is different in diurnal and nocturnal insects. The two types of images formed in ommatidia of the diurnal and nocturnal insects are the **apposition image** and **superposition image**, respectively.

A. Apposition image: These images are formed in diurnal insects like houseflies in which the retinulae lie immediately below the vitrellae and crystalline cone. These are surrounded by secondary pigment cells, which absorb light rays and prevent them from passing to the adjoining ommatidium. In this type, light rays entering the cornea of a single ommatidium only converge on the rhabdome. Thus, a small part of the visual field is focused on to the receptor part in each ommatidium. The total image formed by the compound eye is a **mosaic** of several **small images**. Such an image is called **apposition image** because it is formed by the juxtaposition of small parts of the visual field. This type of vision is called **mosaic vision**.

B. Superposition image: These images are formed in nocturnal insects like **cockroach** in which the retinulae are present deep below the vitrellae and crystalline cone. Retinal sheath is absent. Therefore, the rhabdome and retinulae of an ommatidium receive not only light rays that enter through its own cornea but also light rays that enter through the corneas of the adjoining ommatidia. This results in the overlapping of images. The image formed by overlapping of images is called '**superposition image**'. It is a '*blurred (nor clear) image*'.

II. Ocelli: At the base of each antenna is a fenestra which represents a simple eye or ocellus. Each ocellus consists of a single corneal facet. These are not involved in image formation but are very sensitive to changes in light intensity.